

Smart Energy Meter with GSM Reporting

Verilog metering DSP verified bit-exactly against Python, portable C firmware with a SIM800L AT state machine tested on the host, and slab-tariff billing

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< 0.02 % (6 loads)

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7 = 2.1875 Wh (true 2.44)

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20 / 20 PASS (gcc)

Firmware host tests

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ICE40 HX8K (nextpnr)

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Yosys synth_ice40

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Xilinx xc7 (stat)

Rs 3168 illustrative

Bill (465 units)

VERDICT

SIM PASS BIT-EXACT

FW TESTS PASS SYNTH OK

TIMING MET NO HW RUN

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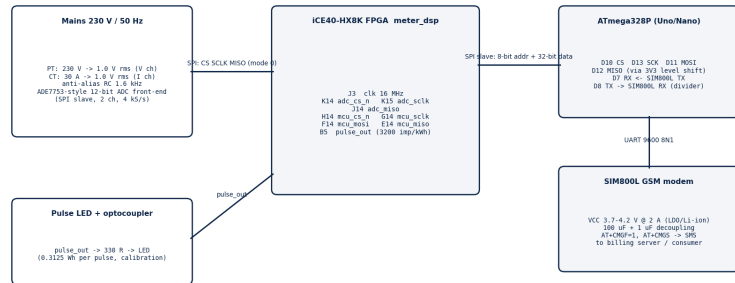
Objective and brief mapping

The brief has three parts: (a) a Verilog metering DSP with a 12-bit ADC sample interface (ADE7753-style SPI front-end modelled in the TB), Vrms, Irms, active power $P = \text{mean}(v.i)$, Wh energy accumulation in a fixed-point MAC, 50 Hz cycle detection and an imp/kWh pulse, verified against a Python reference for several loads and power factors; (b) portable C firmware that reads the block over SPI, formats an SMS with SIM800L AT commands (AT+CMGF=1, AT+CMGS) and handles responses and timeouts with a state machine, built on the host with mock SPI and mock modem, AT transcript saved; (c) a Python billing simulation with an illustrative Indian slab tariff, fixed charge and GST-free electricity duty; plus synth_ice40 + nextpnr stats, a wiring diagram and an Arduino .ino port, with the declaration that no real hardware ran. All parts were built and executed here except the hardware itself.

Architecture

Register (SPI slave, 8-bit addr + 32-bit read)	Content	Scaling used by the firmware
0 VRMS, 1 IRMS	rms x 16 in ADC LSB	$V = \text{reg} / 16 \times 0.2 \text{ V}$, $A = \text{reg} / 16 \times 0.02 \text{ A}$
2 P	signed 24-bit mean(v.i) in LSB^2	$W = \text{reg} \times 0.004$
3 ENERGY	pulse count, 3200 imp/kWh	$\text{kWh} = \text{reg} / 3200$ (0.3125 Wh per pulse)
4 PERIOD16	samples in 16 line cycles	$f = 16 \times 4000 / \text{reg}$ (ADE7753 PERIOD style, no FPGA divider)
5 STATUS	{windows[15:0], ready}	firmware polls ready before converting

Smart energy meter with GSM reporting -- wiring diagram



Declared: no ADE7753, FPGA board, Atmega328P or SIM800L was used; the FPGA block ran in Icarus Verilog, the firmware ran on a PC against mock SPI/GSM.

Implementation details

- meter_dsp.v: SPI master pulls 32 bits (V then I, offset binary) every 250 us; 48-bit sums of v^2 , i^2 , $v.i$ over 4096 samples (1.024 s); $P = \text{sum} \gg 12$; $V_{\text{rms}} \times 16 = \text{isqrt}(\text{sum} \gg 4)$ with a 16-cycle restoring square root (two in parallel); energy MAC adds $v.i$ (only when positive) each sample and pulses when the accumulator passes 1.125e9 units = 0.3125 Wh; cycle detector counts samples over 16 positive zero crossings with +/-12.8 V hysteresis.
- tb/gen_stimulus.py: six scenarios (230 V 5 A PF 1; 230 V 10 A PF 0.8 lag; 235 V 2 A PF 0.5 at 49.8 Hz; 225 V no load; 230 V 8.7 A PF 0.95 with 5 % 3rd harmonic; 230 V 15 A) quantised exactly like the ADC; expected registers computed with the same integer arithmetic (bit-exact), plus true float values for the accuracy table.
- firmware/meter_fw.c: meter_read (6 SPI reads), meter_convert (V, A, W, VA, PF, Hz, kWh), meter_format_sms (<= 160 GSM-7 chars), SIM800L state machine INIT_AT -> INIT_ATE0 -> INIT_CMGF -> READY -> CMGS_WAIT_PROMPT -> CMGS_WAIT_RESULT -> DONE/ERROR with 2 s / 5 s / 60 s timeouts and 3 attempts (ESC cancels a half-typed SMS).
- firmware/host/mock_hal.c: mock SPI register image, behavioural SIM800L (prompt and +CMGS latencies, fault modes: no prompt once, +CMS ERROR once, dead), virtual clock, transcript writer. firmware/arduino/smart_meter_gsm.ino: SPI.h + SoftwareSerial HAL around the same C file.
- billing/: tariff.py (illustrative UP/Delhi-style slabs 5.50 / 5.50 / 6.00 / 6.50 Rs per kWh, fixed 110 Rs/kWh, 5 % electricity duty, no GST), gen_kwh_log.py (30-day household pulse log in 0.3125 Wh steps), billing.py (bill.txt/json/png), test_billing.py.

Verification of the DSP (Icarus Verilog 14, 37 / 37 checks): measured vs true values

All 37 register reads over the MCU SPI port matched the Python reference exactly (tolerance was +/-1 LSB). Power error is below 0.02 % for every load, Vrms within 0.13 % (the 1.024 s window is not a whole number of cycles), frequency 49.767 Hz for a 49.8 Hz input (PERIOD16 resolution). True energy over the 6.14 s run is 2.4399 Wh; the meter emitted 7 pulses = 2.1875 Wh with 0.25 Wh left in the accumulator, i.e. exact within one 0.3125 Wh quantum.

Window / load	Vrms meas / true	Irms meas / true	P meas / true (W)	f (Hz)	Pulses
1: 230 V 5 A PF 1.0	229.86 / 230.0	4.996 / 5.00	1148.75 / 1148.70	50.000	1
2: 230 V 10 A PF 0.8 lag	230.29 / 230.0	10.010 / 10.00	1846.84 / 1846.78	50.000	2

KEY TAKEAWAY Fixing the LSB weights so one LSB^2-sample is exactly 1 uJ turns the energy meter into adders and a compare; RTL, Python model and the C firmware agree to the LSB and the SMS path survives modem timeouts and errors.

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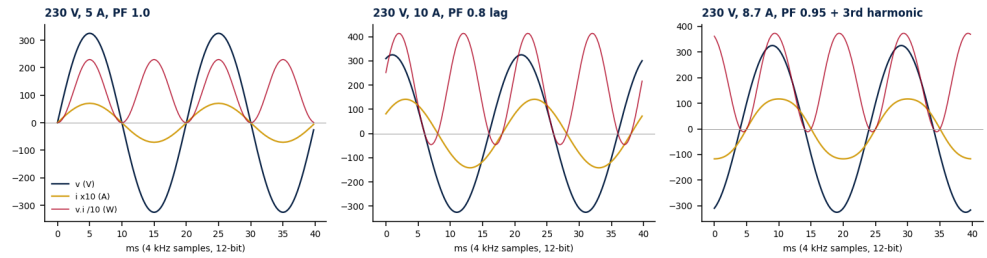
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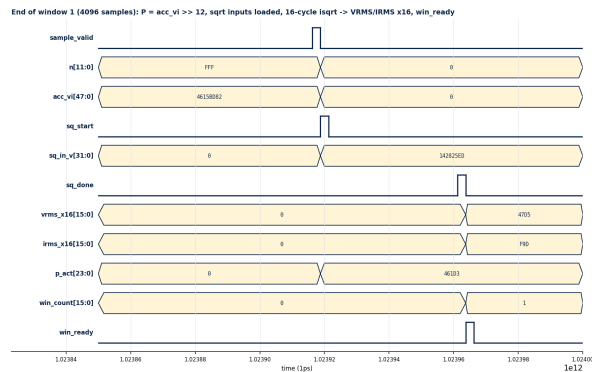
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Window / load	Vrms meas / true	Irms meas / true	P meas / true (W)	f (Hz)	Pulses
3: 235 V 2 A PF 0.5, 49.8 Hz	235.01 / 235.0	1.999 / 2.00	234.98 / 235.01	49.767	3
4: 225 V no load	225.25 / 225.0	0.000 / 0.00	0.00 / 0.00	50.000	3
5: 230 V 8.7 A PF 0.95 + 3rd harm.	229.91 / 230.0	8.714 / 8.70	1901.04 / 1901.32	50.000	4
6: 230 V 15 A PF 1.0	229.86 / 230.0	14.991 / 15.00	3446.16 / 3446.09	50.000	7



The quantised 4 kS/s sample stream the RTL processed (two cycles each of scenarios 1, 2 and 5) with instantaneous power v.i.

Window end and register read-out



End of window 1: on the 4096th sample $P = acc_vi \gg 12 = 0x461D3$ (1148.75 W), sqrt inputs load, isqrt finishes 16 cycles later: $VRMS \times 16 = 0x47D5$ (229.86 V), $IRMS \times 16 = 0xF9D$, win_ready.

Firmware host tests (gcc -std=c11 -Wall -Wextra, mock SPI + mock SIM800L, 20 / 20 PASS)

```
[ 1000 ms] MCU->SIM800L: AT+CMGS="+919360548243"<CR>
[ 1300 ms] SIM800L->MCU: >
[ 1300 ms] MCU->SIM800L: METER EM-0421 kWh=0.001 V=230.3 I=10.01 P=1847W PF=0.80 f=50.00 Est=Rs 0
[ 3800 ms] SIM800L->MCU: +CMGS: 10
[ 3820 ms] SIM800L->MCU: OK
---- scenario B: no '>' prompt on first attempt (5 s timeout), retry ----
[ 5000 ms] MCU->SIM800L: AT+CMGS="+919360548243"<CR>
[ 10000 ms] LOG gsm: no '>' prompt in CMGS_WAIT_PROMPT (retry 1)
[ 10000 ms] MCU->SIM800L: <ESC>
```

Scenario	Behaviour verified	Result
Register conversion (RTL window 2 image)	230.29 V, 10.010 A, 1846.84 W, PF 0.80, 50.00 Hz, 0.000625 kWh; 24-bit P sign extension (-256)	PASS
SMS format	METER EM-0421 kWh=0.001 V=230.3 I=10.01 P=1847W PF=0.80 f=50.00 Est=Rs 0.00 (75 chars <= 160)	PASS
A: init + normal send	AT / ATE0 / AT+CMGF=1 -> READY; AT+CMGS="+91..." -> '>' -> text + Ctrl-Z -> +CMGS: 10 -> OK -> DONE; modem text identical	PASS
B: no '>' prompt	5 s timeout -> ESC + retry -> DONE, retries = 1	PASS
C: +CMS ERROR: 500	retry -> DONE (3 SMS delivered in total)	PASS
D: dead modem	3 x 5 s timeouts -> ERROR (result -1); new SMS accepted; recovers when the modem answers	PASS

Billing simulation (illustrative UP/Delhi-style domestic slab tariff -- not an official rate card)

Line item	Units / basis	Rate	Amount (Rs)
Slab 1-100	100 units	5.50 / kWh	550.00
Slab 101-150	50 units	5.50 / kWh	275.00
Slab 151-300	150 units	6.00 / kWh	900.00
Slab > 300	165 units	6.50 / kWh	1072.50
Energy charges	465 units (464.968 kWh; reading 42 970 -> 1 487 898 pulses)		2797.50

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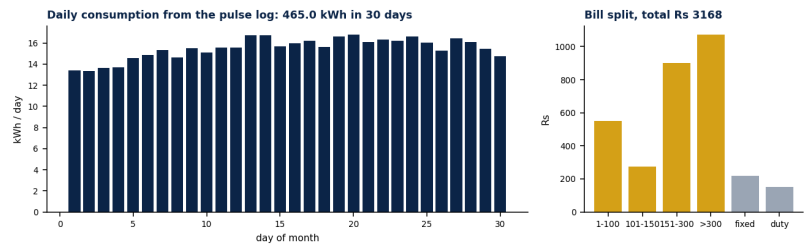
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Line item	Units / basis	Rate	Amount (Rs)
Fixed charge	2.0 kW sanctioned	110 / kW / month	220.00
Electricity duty (electricity is outside GST)	on energy + fixed	5 %	150.88
TOTAL PAYABLE (rounded)	period 2026-05-01 .. 2026-05-30		3168.00

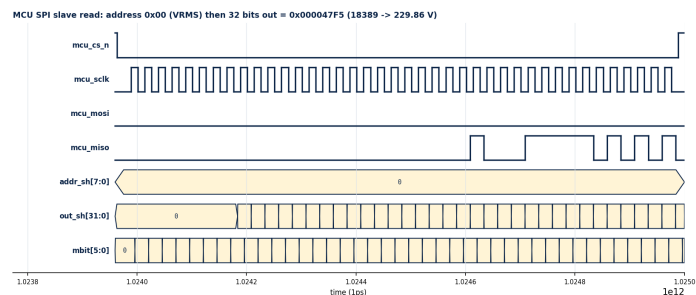


Daily kWh from the simulated 30-day pulse log and the bill split; billing/test_billing.py 3/3 unittest cases pass.

Synthesis of the DSP block (real configuration: 16 MHz, 4 kS/s, 4096-sample window)

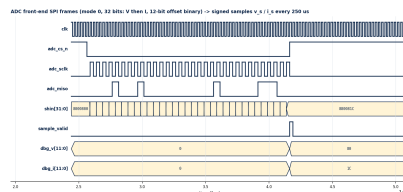
Flow (run here)	Measured result
Yosys 0.68 synth_ice40	3323 cells: 2022 SB_LUT4, 684 FF, 616 SB_CARRY (three 12x12 multipliers and 48-bit accumulators in LUT/carry logic: HX8K has no DSP blocks)
nextpnr-ice40 hx8k ct256, --freq 16 --seed 1	2393 / 7680 LC (31 %), 156 SB_IO, 7 SB_GB; Fmax 44.69 MHz, PASS at 16 MHz
icetime -d hx8k	total path delay 22.87 ns (43.72 MHz)
Yosys synth_xilinx xc7 (stat)	1729 cells: 769 LUT, 612 FF, 125 CARRY4, 3 DSP48E1

MCU register read



MCU side (mode 0): 8 address bits on MOSI (addr_sh = 0x00), out_sh loaded on the 8th falling edge, then 32 bits on MISO = 0x000047F5 = 18389 -> 229.86 V.

ADC front-end frames



ADC side: CS low, 32 SCLK edges, MISO bits shifted into shin; on the last bit v_s / i_s update (offset binary -> two's complement) and sample_valid pulses, every 250 us.

Declared substitutions / not run here

- No ADE7753 / ADC front-end, FPGA board, ATmega328P or SIM800L, no SIM card or network: the ADC is a Verilog SPI-slave model fed with Python-synthesised waveforms, the modem is a C behavioural model and the AT transcript comes from it.
- RTL simulated with Icarus Verilog at a sim-scaled 400 kHz clock (100 clocks per sample); synthesis at the real 16 MHz; the nextpnr bitstream (regenerated by make synth, not kept in the repo) was not programmed. The Arduino sketch was not compiled (no avr-gcc here). Xilinx numbers are Yosys estimates, no Vivado.
- Tariff and 30-day consumption are illustrative simulations, clearly labelled, not DISCOM rates or a real reading.

Learnings / next steps

Picking 0.2 V and 0.02 A LSBs at 4 kS/s made one LSB²-sample exactly 1 uJ, so the energy path is an adder and an integer compare and the Python model matches the RTL bit for bit. A bit-exact reference caught the one-sample misalignment of my first SPI slave model immediately. Next: reactive power and PF inside the FPGA, calibration registers, CT tamper detection and a real SIM800L bring-up with the .ino.

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